

Final Exam Study Guide

01- INTRO TO COGNITIVE PSYCHOLOGY

1. What is Cognitive Psychology?
2. How do we study Cognition?
 - a. Experimental Psychology
 - b. Neuropsychology
3. Describe the early work in Cognitive Psychology
 - a. Donders
 - i. Simple RT
 - ii. Choice RT
 - b. Ebbinghaus
 - i. Forgetting curve
 - ii. savings
 - c. Wundt
 - i. Structuralism
 - ii. Analytic introspection
 - d. James
 - i. Firsts?
4. Describe the early work done by behaviorists which set the stage for the Cognitive Revolution.
 - a. Tolman
 - i. Cognitive Map
 - b. Other behaviorists- what did they think about studying cognition?
5. What is the “Cognitive Revolution” and how was it impacted by the advent of the computer?
 - a. Began in 1950s
 - b. Gradual
 - c. Information processing approach/models

02-INTRO TO COGNITIVE NEUROSCIENCE

1. What are the main parts of a neuron?
 - a. Soma
 - b. Dendrites
 - c. Axon
 - d. Other aspects?
2. How is information transferred in the nervous system?
 - a. Neural Firing
 - i. Neurons receive a signal that is in either a chemical form or a physical form.
 - ii. Sodium diffuses down through the dendrites and cell body.
 - iii. The electrical signal is activated and travels quickly down the axon to the axon terminals.

1. Steps in an Action Potential
 - a. Where?
 2. Myelination- why?
 3. All or None Law
 - iv. This causes the release of neurotransmitters at the synapse and the process starts all over in the next neuron.
3. How is information represented in the nervous system?
 - a. Neural Representation
 - i. Feature Detectors
 - ii. Stimulus Coding
 1. Specificity Coding
 2. Population Coding
 3. Sparse Coding
4. What are some of the ways we can study neural communication & what are the pros & cons of each?
 - a. Neuronal Recording
 - i. Single Cell Recording
 1. Receptive Fields
 2. Pros and Cons?
 - ii. EEG/ERPs
 1. Pros and Cons?
 - b. Spatial Localization
 - i. Single Cell Recording
 - ii. Neuropsychology
 - iii. MRI
 - iv. fMRI
 - v. Pros and Cons of each method?
5. What are some of the important anatomical landmarks in the brain?
 - a. Basic Anatomy & Localization of Function
 - i. Lobes- Functions?
 - ii. Distributed Representation
 1. Structural Connectivity
 2. Functional Connectivity
 - a. Common Neural networks determined by resting-state fMRI
6. What are some of the important anatomical locations for some of our main cognitive functions?
 - a. Related to above question ^
 - b. Perception- receiving areas for the senses
 - c. Frontal Lobe
 - d. Broca's Area
 - e. Wernicke's Area
 - f. FFA

03 -PERCEPTION

1. What is the difference between Sensation and Perception?
 - a. Bottom-up versus top-down processing
2. Give some examples of how knowledge and previous experience play a role in forming our perceptions.
 - a. Where/How does knowledge and previous experience come in?
 - i. Object Recognition
 - ii. Speech Segmentation
 - iii. Pain Perception
3. Compare 4 theories of how we perceive/recognize objects
 - a. Helmholtz's unconscious inference
 - b. Gestalt principles of organization
 - i. Gestalt Laws- each one that we discussed
 - c. Considering Environmental Regularities
 - i. Physical regularities
 - ii. Semantic regularities
 - d. Bayesian Inference
 - e. Compare the 4
4. Explain plasticity in the perceptual system.
 - a. Vertical vs Horizontal lines
 - i. Selective Rearing
 - b. FFA- can you train FFA to respond to new stimuli?
 - i. Greebles
 - c. Somatosensory cortex
 - i. Homunculus
 - ii. Monkey studies
 - iii. Training/practice
 - iv. Phantom limb
5. What is the evidence from animal lesion studies and from Neuropsychology that there are 2 separate perceptual pathways? Compare and contrast the location and function of these pathways.
 - a. What vs Where pathways
 - i. What lobes?
 - ii. Monkey lesion study
 1. Object Discrimination
 2. Landmark discrimination
 3. Double dissociation
 - iii. Patient DF
 1. Agnosia
 2. Prosopagnosia
 3. Versus RV?

10 -VISUAL IMAGERY

1. Describe evidence for and against the argument that imagery and perception involve the same mechanisms.
 - a. Imagery and Perception
 - i. Spatial Correspondence
 1. Shepard & Metzler rotation tasks
 - a. Mental chronometry
 2. Mental Scanning task
 - a. Map Task
 3. Mental walks
 - ii. DEBATE- Spatial vs. Propositional
 - a. Pylyshyn- Not Spatial but propositional
 - b. Imagery works like perception- must be spatial
 - i. Viewing distance study
 - c. Interactions
 - i. Dim projection study
 - ii. Farah- Priming by imagery study
2. Describe similarities and differences in the brain mechanisms involved in imagery and perception
 - a. Kreiman et al
 - b. Le Bihan et al 1993
 - c. Ganis et al 2004
3. Describe how method of loci and Pegword techniques use imagery to improve memory.
 - a. Method of loci
 - b. Pegword technique

04- ATTENTION

1. Describe Inattentional Blindness and give some examples
 - a. Inattentional Blindness
 - b. Change Blindness
2. Compare and contrast the three main models of attention (Broadbent, Treisman, and MacKay).
 - a. Early
 - i. Broadbent's Early Selection Theory
 - ii. Cocktail Party Phenomenon
 - iii. Cherry's Shadowing Experiments
 - b. Attenuation
 - i. Treisman's Attenuation theory
 1. Dictionary Unit
 - c. Late
 - i. McKay
3. Describe the Stroop task and the Attentional Blink task and their typical results. What do these tasks demonstrate about attention?
 - a. Stroop Task
 - b. Attentional Blink Paradigm

- c. Inattentional Deafness
- 4. Compare and contrast Overt and Covert Attention and give examples of each.
 - a. Overt vs. Covert
 - i. Overt
 - 1. Eye-tracking
 - ii. Covert
 - 1. Helmholtz
 - 2. Precuing
- 5. Compare and contrast Automatic (Exogenous) and Voluntary (Endogenous) attention.
 - a. Voluntary
 - i. Completing tasks
 - ii. Scene Schema
 - iii. Divided attention
 - b. Reflexive
 - i. Saliency maps
- 6. What are some of the physical and meaningful things that can automatically affect the allocation of our attention?
 - a. Physical:
 - i. Salience
 - 1. Movement
 - 2. Color
 - b. Meaningful:
 - i. Faces
 - ii. Schemas
 - iii. Memory and Familiarity
 - iv. Addictive Stimuli

05 -STM & WM

- 1. Explain how information is processed in Atkinson and Shiffrin's modal model of memory.
 - a. Sensory Memory
 - b. STM
 - i. Control Processes?
 - c. Long-Term Memory
- 2. Describe evidence demonstrating the large capacity and rapid decay of sensory memory.
 - a. Sensory Memory
 - i. Echoic
 - 1. Duration?
 - ii. Iconic
 - 1. Capacity & Duration?
 - a. Partial Report task
- 3. Describe evidence demonstrating the limited capacity and relatively brief duration of short-term memory.

- a. Duration?
 - i. Proactive Interference
- b. Capacity?
 - i. Miller says?
 - ii. But- visual/more complicated stimuli?
 - iii. Chunking
 - iv. Items or amount of info?
- 4. Contrast Baddeley's concept of working memory with Atkinson and Shiffrin's description of the short-term memory system.
 - a. WM versus STM?
- 5. Describe the functions of the components of Baddeley's revised working memory model.
 - a. WM- 3 parts (+ one added later!)
 - i. Phonological Loop
 - ii. Visuospatial sketch pad
 - iii. Central Executive
 - iv. Episodic Buffer
- 6. Describe evidence in humans and monkeys suggesting that the prefrontal cortex plays an important role in working memory

06 -LTM- STRUCTURE

- 1. Explain how research on the serial position effect has been used to distinguish between short-term and long-term memory processes.
 - a. LTM- Relationship with STM
 - i. Serial Position Curve
 - 1. Primacy Effect- Why?
 - 2. Recency Effect-Why?
 - 3. What happens when you introduce a delay?
 - 4. Proactive Interference?
- 2. Provide examples of visual, auditory, and semantic coding in both short-term and long-term memory.
 - a. Compare Coding in STM and LTM
 - i. Visual
 - ii. Auditory
 - iii. Semantic
 - iv. Which is most common in STM? In LTM?
- 3. Describe evidence suggesting that short-term and long-term memory depend on separate brain regions.
 - a. Localization of Memory in the Brain
 - i. Amnesia- Retrograde and Anterograde
 - 1. Define each
 - 2. HM & Clive Wearing
 - a. Hippocampus
 - b. versus KF?

- c. Double Dissociation
- 4. Describe the ways in which episodic and semantic memory are distinct from one another and the ways in which they interact.
 - a. Explicit Memory (Conscious)
 - i. Episodic
 - 1. Mental time travel
 - ii. Semantic
 - 1. Knowledge
 - iii. Evidence they are separate
 - 1. Neuropsychological Case Studies- KC and LP
 - 2. fMRI research
 - 3. But, they interact? Autobiographical memories
 - 4. Effect of time- Semanticization of memory
- 5. Describe research findings suggesting that the same brain regions are involved in representing episodic memory for the past and in the ability to imagine future events.
 - a. Constructive episodic simulation hypothesis
- 6. Describe differences between implicit memory, including procedural memory, priming, and classical conditioning.
 - a. Implicit Memory
 - i. Procedural
 - ii. Priming
 - 1. Perceptual
 - 2. Propaganda effect
 - 3. Real life priming?
 - iii. Classical Conditioning
 - iv. Habituation and Sensitization

Do people with amnesia still retain implicit memory? Learn new skills? .

Analyze- Memory in the Movies

- Task from in class (posted on Canvas under in-class activities)

07 -LTM- ENCODING & RETRIEVAL

- 1. What are some research proven strategies to increase your retention of the material in this class for the exams and beyond?
 - a. Consolidation
 - i. Amnesia pattern
 - ii. Sleep effect
 - b. Encoding
 - i. Rehearsal
 - 1. Maintenance
 - 2. Elaborative

- a. Levels of Processing Theory
 - i. Shallow vs Deep
 - 1. Physical
 - 2. Rhyming
 - 3. Meaning
 - b. Circular reasoning
 - c. “Deep”- or better ways of encoding:
 - i. Organization & Categorization
 - ii. Mnemonics
 - iii. Complex Sentences
 - iv. Visual Imagery
 - 1. Method of Loci
 - 2. Peg word technique
 - v. Personal connection
 - vi. Generation
 - vii. Testing effect (Retrieval Practice)
 - viii. Distributed Practice
 - c. Provide examples of these strategies.
- 2. Describe the importance of retrieval cues as revealed by differences in memory performance in tests of free recall compared to cued recall.
 - a. Most of our failures of memory are failures to retrieve
 - b. Retrieval cues
 - 3. Explain how encoding specificity, state-dependent learning, and transfer-appropriate processing can improve memory retrieval.
 - a. Transfer Appropriate processing
 - b. Context Cues
 - i. Encoding Specificity
 - ii. State-dependent learning

08- EVERYDAY MEMORY & MEMORY ERRORS

- 1. Describe similarities and differences between flashbulb memories and other more ordinary or “everyday” autobiographical memories
 - a. Forgetting can be good?
 - i. Hyperthymesia
 - ii. Patient AJ (Jill Price)
 - b. Memory for Emotional Stimuli
 - i. Flashbulb Memories
 - 1. Are they better remembered?
 - 2. How do they differ from other memories?
 - 3. Narrative Rehearsal Hypothesis
- 2. Explain how the constructive nature of memory is revealed by research
 - a. Memory is a construction- not a photograph or video recording

- i. Source monitoring
 - 1. Source monitoring error
 - a. Becoming famous overnight
 - b. Illusory truth effect
 - 2. Cryptomnesia
 - 3. Schemas and Scripts
 - ii. Advantages and Disadvantages of constructive nature of memory?
 - iii. Power of Suggestion- Misinformation Effect
 - 1. MPI
 - 2. Stop sign study
 - 3. Smashed vs hit study
 - 4. Lindsay 1990 study
 - iv. DRM procedure
 - v. Implanting Childhood memories
3. Explain a few reasons why we need to be cautious of eyewitness testimony
- a. Errors in Eyewitness Testimony
 - i. Assumption Perps in lineup
 - ii. Errors due to attention
 - iii. Errors due to familiarity
 - iv. Errors due to suggestion
 - v. What are they doing about it?
 - 1. Line-up Best practices
 - 2. Interview Best Practices

11- LANGUAGE

- 1. What are the levels of the hierarchy for language- describe each?
 - 1. Human Language- Universal and Hierarchical
 - a. Language Rules-Syntax vs Semantics
 - b. Levels of Language Processing
 - i. Phonemes
 - ii. Morphemes
 - iii. Words
 - iv. Sentences
 - v. Text and Stories
- 2. What are some of the ways top-down processing helps us perceive and understand phonemes (and letters in written language)?
 - a. Phonemes
 - i. Production
 - ii. Perception (Top-down + bottom up processing)
 - 1. McGurk Effect
 - 2. Word-superiority (written words)

3. Phonemes in words
4. Phonemic Restoration Effect
3. Describe the mental lexicon and how it is thought to be arranged.
 - a. Morphemes
 - b. Lexical “neighborhoods”
 - c. Word Frequency
4. What are some complications in understanding words and how do we resolve them?
 - a. Word Frequency effect
 - b. Variable pronunciation
 - c. Segmentation
 - i. Why is it difficult?
 - ii. Context
 - d. Lexical Ambiguity
 - i. Meaning dominance
 - ii. Context
5. Describe the differences between Broca’s and Wernicke’s aphasics.
 - a. Semantics versus syntax
 - b. Parsing
 - c. Agrammatism
 - d. Anomia
 - e. Speech production vs language comprehension
 - f. Areas of the brain?
6. What is common ground and how is it established in a conversation?
 - a. Common Ground
 - i. Entrainment
 - ii. Syntactic Coordination
 1. Syntactic priming
 - iii. Other
 1. Theory of Mind
 2. Nonverbal Communication

13- JUDGEMENT, REASONING, & DECISION MAKING

1. Explain how the availability heuristic, illusory correlations, and the representative heuristic can cause errors in reasoning.
 - a. Are you able to define each and give examples?
2. Explain how the myside bias, confirmation bias, and limitations to our ability to objectively evaluate evidence can influence our judgments.
 - a. Are you able to define each and give examples?
3. Describe research suggesting that people often fail to follow the decision-making procedures proposed by expected utility theory.
 - a. Expected Utility Theory
 - i. Exceptions
 1. Eg. Deal or no deal
 2. Examples from game

- ii. Why irrational?
 - 1. Emotion role in decisions
 - a. Expected
 - b. Immediate
 - i. Integral
 - ii. Incidental
- 4. Explain how decision making can be influenced by emotions, contextual factors, and the way that choices are presented.
 - a. How choices presented/context
 - i. Status Quo Bias
 - ii. Framing effect
 - 1. Risk aversion
 - 2. Risk-taking
 - b. Decision Fatigue